



PAPER ID-311016

Printed Page: 1 of 2
Subject Code: RCA1501

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MCA-INT
(SEM V) THEORY EXAMINATION 2024-25
DESIGN & ANALYSIS OF ALGORITHMS

TIME: 3 HRS**M.MARKS: 70****Note:** Attempt all Sections. In case of any missing data; choose suitably.**SECTION A****1. Attempt all questions in brief.****2 x 07 = 14**

Q no.	Question	CO	Level
a.	Define asymptotic behavior of an algorithm.	1	1
b.	What is the significance of balancing in B-trees?	2	2
c.	Explain the different types of relations	2	2
d.	What is the Minimum Cost Spanning Tree problem? Explain the Greedy approach for finding it.	3	2
e.	Explain the concept of Eulerian Path and Eulerian Circuit in graphs.	3	2
f.	What is the Sum of Subsets problem?	4	1
g.	What is the significance of approximation algorithms in optimization problems?	5	2

SECTION B**2. Attempt any three of the following:****07 x 3 = 21**

Attempt any three of the following:																		
a.	For the given function: $T(n) = 3n^2 + 5n + 7$. Determine the Big O, Big Omega, and Big Theta notations for the time complexity of the function.	1	3															
b.	Given the in-order and post-order traversals of a binary tree, reconstruct the binary tree. Consider the following: In-order traversal: M, N, O, P, Q, R, A, B, C, D, E, F, G, H, I, J, K, L Post-order traversal: M, O, N, P, Q, R, B, D, E, G, F, H, K, L, J, I, C, A	2	3															
c.	Consider the following items with weights and values: <table border="1"><thead><tr><th>Item</th><th>Weight</th><th>Value</th></tr></thead><tbody><tr><td>1</td><td>2</td><td>40</td></tr><tr><td>2</td><td>3</td><td>50</td></tr><tr><td>3</td><td>4</td><td>70</td></tr><tr><td>4</td><td>5</td><td>60</td></tr></tbody></table> Given the knapsack capacity = 5 kg, apply the Greedy method to solve the 0/1 Knapsack problem.	Item	Weight	Value	1	2	40	2	3	50	3	4	70	4	5	60	3	3
Item	Weight	Value																
1	2	40																
2	3	50																
3	4	70																
4	5	60																
d.	Explain the graph coloring problem. How can backtracking be used to color a graph?	4	2															
e.	Explain why the Halting Problem is considered NP-Hard but not NP-Complete.	5	2															

SECTION C**3. Attempt any one part of the following:****07 x 1 = 07**

a.	Compare the asymptotic behavior of different sorting algorithms like Quick Sort, Merge Sort, and Bubble Sort in terms of best, average, and worst-case scenarios.	1	2
b.	Compare and contrast polynomial, logarithmic, and exponential time complexities with suitable examples.	1	2

4. Attempt any one part of the following:**07 x 1 = 07**

a.	Perform an insertion operation in a B-tree of order 4. Given the following sequence of keys to insert: [50, 20, 30, 40, 60, 10, 70, 80, 90, 15, 35, 25]	2	3
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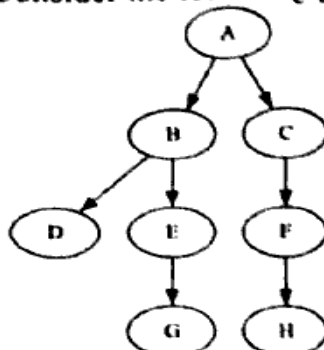
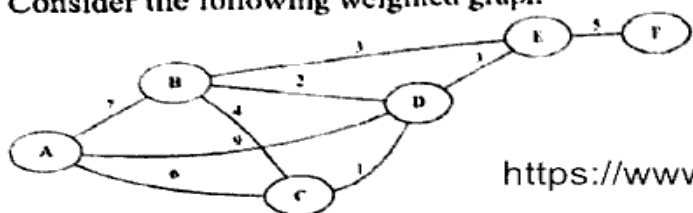
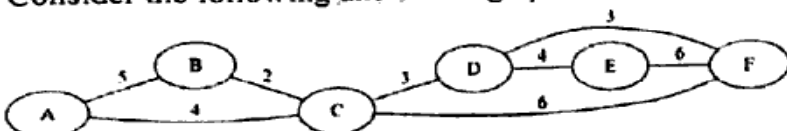
PAPER ID-311016

Roll No: Printed Page: 1
Subject Code: RCAI

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M.MARKS: 7

TIME: 3 HRS

b.	Consider the following graph:  Perform Depth First Search (DFS) and Breadth First Search (BFS) starting from node A.	2	3																									
5. Attempt any one part of the following:		07 x 1 = 07																										
a.	Consider the following weighted graph  https://www.aktuonline.com What is the shortest path from A to F, and what is the total weight of the path?	3	3																									
b.	Consider the following undirected graph:  Find out if the graph has an Eulerian Path. If it does, show the path, or explain why it does not exist.	3	3																									
6. Attempt any one part of the following:		07 x 1 = 07																										
a.	Explain the difference between backtracking and branch and bound with examples.	4	3																									
b.	Solve the Travelling Salesman Problem (TSP) using branch and bound. Given the following distances between cities, find the minimum cost of visiting all cities. Consider the following distance Matrix: <table border="1" data-bbox="327 1637 1327 1823"><thead><tr><th>Cities</th><th>A</th><th>B</th><th>C</th><th>D</th></tr></thead><tbody><tr><th>A</th><td>0</td><td>10</td><td>15</td><td>20</td></tr><tr><th>B</th><td>10</td><td>0</td><td>35</td><td>25</td></tr><tr><th>C</th><td>15</td><td>35</td><td>0</td><td>30</td></tr><tr><th>D</th><td>20</td><td>25</td><td>30</td><td>0</td></tr></tbody></table> Solve using the branch and bound technique to find the least cost tour.	Cities	A	B	C	D	A	0	10	15	20	B	10	0	35	25	C	15	35	0	30	D	20	25	30	0	4	3
Cities	A	B	C	D																								
A	0	10	15	20																								
B	10	0	35	25																								
C	15	35	0	30																								
D	20	25	30	0																								
7. Attempt any one part of the following:		07 x 1 = 07																										
a.	Explain the implications if $P = NP$ or $P \neq NP$.	5	3																									
b.	Discuss the significance of Cook's Theorem in the context of NP-completeness.	5	3																									